

Diploma Supplement

University of
Stavanger



This Diploma Supplement model was developed by the European Commission, Council of Europe and UNESCO/CEPES. The purpose of the supplement is to provide sufficient independent data to improve the international 'transparency' and fair academic and professional recognition of qualification (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It should be free from any value judgements, equivalence statements or suggestions about recognition. Information in all eight sections should be provided. Where information is not provided, an explanation should give the reason why.

1 INFORMATION IDENTIFYING THE HOLDER OF THE QUALIFICATION

- 1.1 Family name(s): Erlandsen
1.2 Given name(s): Vebjørn
1.3 Date of birth (day/month/year): 28-05-2004
1.4 Student identification number or code: 277963

2 INFORMATION IDENTIFYING THE QUALIFICATION

- 2.1 Name of qualification and (if applicable) title conferred (in original language):
Bachelor i ingeniørfag
The title Bachelor is protected by law in Norway.
- 2.2 Main field(s) of study for the qualification:
Electrical Engineering
Mathematics and Statistics
Natural Sciences (Physics, Chemistry and Computer Science)
- 2.3 Name and status of awarding institution (in original language)
Universitetet i Stavanger, a public university. The quality assurance system was evaluated and approved by the Norwegian Agency for Quality Assurance in Education in 2021.
- 2.4 Name and status of institution administering studies
See section 2.3
- 2.5 Language(s) of instruction/examination
Norwegian

3 INFORMATION ON THE LEVEL OF THE QUALIFICATION

- 3.1 Level of qualification
First Cycle/Level 6, Norwegian Qualifications Framework for Lifelong Learning

- 3.2 Official length of the programme
3 years in full-time mode (180 ECTS credits)
- 3.3 Access requirements
Higher Education Entrance Qualification with specific entrance requirements

4 INFORMATION ON THE CONTENTS AND RESULTS GAINED

- 4.1 Mode of study
Full-time
- 4.2 Programme requirements

Objectives, Content and Organisation of the Programme of Study

The bachelor's programme in Control Engineering and Circuit Design provides students with a broad engineering foundation in electrical engineering, with specialisation in automation and electronics design. Graduates acquire solid competence in mathematics, physics and programming, as well as specialised knowledge of analogue and digital electronics, control engineering and system-level understanding. The programme emphasises the ability to develop innovative and sustainable solutions through theoretical analysis and practical laboratory work.

The programme qualifies graduates for professional practice as electrical engineers across a wide range of industries, including the power sector, process industry, telecommunications and technology companies. Graduates are well prepared for roles involving the design, development and operation of complex control systems, electronics and renewable energy technologies.

To be awarded the degree Bachelor of Engineering, the candidate must complete at least 180 ECTS credits, comprising of the following:

- Engineering basis: 30 ECTS credits in basic mathematics, engineering systems thinking, and an introduction to professional engineering practice and methods. This should primarily relate to the engineering programme and form the foundation of the engineering programme.
- Programme basis: 50–70 ECTS credits in technical courses, natural sciences, and social sciences. This should primarily relate to the study programme and form the foundation of the field.
- Technical specialisation: 50–70 ECTS credits that provide a clear direction within the student's own field of study, and which build on the engineering foundation and the programme-specific foundation. This should primarily relate to the chosen specialisation and form the basis for the field of study.
- Elective courses: 20-30 ECTS credits that contribute to further academic specialisation, either in scope or in depth.

Elective courses are generally taken towards the end of the programme, where a period of study abroad is also possible.

The bachelor's thesis is a final project of 20 ECTS credits and is mandatory for all students. It may be conducted individually or in groups and forms part of the technical specialisation. The thesis must address real-world problems from industry, public sector, or research and development, and includes an introduction to research methodology and the theory of science.

Students encounter varied teaching and assessment methods, the use of modern digital tools, practical laboratory work, and project-based assignments involving report writing and documentation. Details on teaching and learning methods, required literature, evaluation methods, and assessment criteria are provided in each course description.

The programme has been completed in accordance with the National Curriculum Regulations for Engineering Education, laid down by the Ministry of Education and Research on 18 May 2018.

The candidate's learning outcomes

A candidate who has completed his or her qualification should have the following learning outcomes defined in terms of knowledge, skills and general competence:

Knowledge

K1: The candidate has broad knowledge that provides a holistic systems perspective on engineering in general, with a specialisation in Electrical Engineering. The candidate has knowledge of electric and magnetic fields, analogue and digital electronics, computer architecture, instrumentation systems, control engineering, signal processing and robotics.

K2: The candidate has foundational knowledge in mathematics, physics, statistics, chemistry, programming and relevant social and economic subjects, and of how these can be integrated into electrical engineering problem-solving.

K3: The candidate has knowledge of the engineer's role in society, as well as how technological developments within electrical engineering have shaped modern society.

K4: The candidate is familiar with research and development work within engineering, with particular regard to the application of electronics and methods within automation, instrumentation, signal processing and robotics.

K5: The candidate is able to update their knowledge within electrical engineering, both through gathering information and through familiarity with professional communities and practice.

Skills

S1: The candidate is able to apply knowledge and relevant findings from research and development work to solve electrical engineering problems involving modelling, simulation and analysis, and is able to make justified decisions where required.

S2: The candidate has professional digital competence in programming (Python, Matlab, Simulink, LabView, PLC, C).

S3: The candidate has practical competence in the use of oscilloscopes, multimeters, signal generators and the connection of equipment, and is proficient in various measurement principles.

S4: The candidate is able to identify, plan and carry out electrical engineering projects, tasks, trials and experiments both independently and as part of a team. The candidate is further able to document this work in report form.

S5: The candidate is able to identify and use electrical engineering information and subject matter to illuminate an electrical engineering problem.

S6: The candidate is able, through innovation and entrepreneurship, to contribute to finding solutions to electrical engineering problems with the aim of creating sustainable and socially beneficial products or systems.

General Competence

G1: The candidate has insight into the environmental, health-related, societal and economic consequences of products and solutions within electrical engineering, and is able to consider these from an ethical and life-cycle perspective.

G2: The candidate is able to identify security, vulnerability, privacy and data protection aspects in products and systems that employ ICT.

G3: The candidate is able to communicate the significance and consequences of technology to a range of audiences.

G4: The candidate can reflect on their own professional practice, including in team and interdisciplinary contexts, and can adapt it to the current work situation.

G5: The candidate can contribute to the development of good practice by participating in professional discussions within the field and sharing their knowledge and experience with others

4.3 Programme details:

Name: Vebjørn Erlandsen

Course		Semester	Credits	Grade	Grade distribution				
					A	B	C	D	E
DAT120	Introduction to Programming	2023 autumn	10	B					
FYS102	Physics for Data/Electro	2023 autumn	5	C					
KJE101	Fundamental Chemistry	2023 autumn	5	B					
MAT100	Mathematical Methods 1	2023 autumn	10	A					
ELE100	Electrical Engineering 1	2024 spring	10	A					
ELE130	Applied Mathematics and Physics in Programming of Robots	2024 spring	10	C					
MAT200	Mathematical Methods 2	2024 spring	10	A					
ELE200	Electrical Engineering 2	2024 autumn	10	A					
ELE210	Computer Architecture	2024 autumn	10	A					
ELE310	Control Technology	2024 autumn	10	B					
ELE300	Signals and Systems	2025 spring	10	B					
ELE320	Control Systems	2025 spring	10	B					
STA100	Probability and Statistics 1	2025 spring	10	A					
ELE340	Electronics and Embedded design	2025 autumn	10	B					
ELE610	Applied Robot Technology	2025 autumn	10	Passed					
MAT300	Vector Analysis	2025 autumn	10	A					
ELEBAC	Bachelor's Thesis in Electrical Engineering <i>Simulation-Based Testing Framework for Embedded Systems Using MQTT and Cross-Compiled Code</i>	2026 spring	20	A					
ING200	Technology Management	2026 spring	10	B					
Total: 180.0									

Credit system and grading

The academic year normally runs from mid-August to mid-June and lasts for 10 months. Courses are measured in "studiepoeng", considered equivalent to the European Credit Transfer System standard (ECTS credits). The full-time workload for one academic year is 1500 - 1800 hours of study / 60 "studiepoeng".

The Norwegian grading system consists of two grading scales: one scale with the grades pass or fail and one graded scale from A to E for pass and F for fail. The graded scale has the following qualitative descriptions:

A	Excellent	An excellent performance, clearly outstanding. The candidate demonstrates excellent judgement and a very high degree of independent thinking.
B	Very good	A very good performance. The candidate demonstrates sound judgement and a high degree of independent thinking.
C	Good	A good performance in most areas. The candidate demonstrates a reasonable degree of judgement and independent thinking in the most important areas.
D	Satisfactory	A satisfactory performance, but with significant shortcomings. The candidate demonstrates a limited degree of judgement and independent thinking.
E	Sufficient	A performance that meets the minimum criteria, but no more. The candidate demonstrates a very limited degree of judgement and independent thinking.
F	Fail	A performance that does not meet the minimum academic criteria. The candidate demonstrates an absence of both judgement and independent thinking.

The assessment is criterion referenced.

Grade distribution

The distribution of grades is shown by the percentage for courses using the graded scale A – F. Fail (F) is not included in the distribution. All results from the last five years are included in the calculation. The distribution is also shown for courses that have been active for less than five years. There has to be at least 10 approved results during the period.

4.4 Grading scheme and, if available, grade distribution guidance:

See section 4.3

See section 4.3

4.5 Overall classification of the qualification (in original language):

Not applicable

5 INFORMATION ON THE FUNCTION OF THE QUALIFICATION

5.1 Access to further study:

The bachelor's degree is at an academic level that is sufficient for application to relevant second cycle studies.

5.2 Professional status:

The qualification is a final award and entitles its holder to exercise professional work within his/her field of study.

6 ADDITIONAL INFORMATION

6.1 Additional information:

Not applicable.

6.2 Further information sources:

University of Stavanger, <http://www.uis.no>

Norwegian Agency for Quality Assurance, <http://www.nokut.no/en>

7 CERTIFICATION OF THE SUPPLEMENT

7.1 Date: 15 June 2026

Date of original qualification: 10 June 2026

7.2 Signature:

The document is electronically signed by Sikt - Norwegian Agency for Shared Services in Education and Research. The document is only valid in its original electronic form with the accompanying electronic signature. Time for signing document 2026-06-15 14:41

7.3 Capacity:

7.4 Official stamp:

8. INFORMATION ON THE NATIONAL HIGHER EDUCATION SYSTEM

Higher education in Norway: Legislature

The Ministry of Education and Research has the overall responsibility for higher education in Norway. Higher education is offered by four types of higher education institutions: university (*universitet*), specialized university institution (*vitenskapelig høyskole*), accredited university college (*akkreditert høyskole*), and university college with accredited study programmes (*høyskole med akkrediterte studier*). The differences between the types of higher education institutions are related to their self-accrediting authority.

All public and private higher education in Norway is subject to the Act Relating to Universities and University Colleges (LOV-2024-03-08-9). An institution's right to award specific degrees and the prescribed lengths of study are codified in Regulation concerning degrees and titles (FOR-2005-12-16 nr 1574). The awarding of master's degrees and requirements for admission to master's degrees is regulated by the Regulations on requirements for awarding a master's degree (FOR-2005-12-01 nr 1392).

Since 2002 Norway has adhered to the objectives of the Bologna Process in the European Higher Education Area. Most of the elements have been implemented through the reform of the Norwegian higher education system carried out in 2003. Central to the reform has been a transition from the former degree system to the bachelor's, master's and doctoral degree structure, with a few exceptions.

Norwegian higher education qualifications make up the levels from 6 to 8 of the Norwegian Qualifications Framework for Lifelong Learning (NQF) from 2011, which is the national overarching qualifications framework¹. It describes the levels of qualifications as defined by the total learning outcomes in terms of the knowledge, skills and general competence that graduates at various levels should have achieved².

NQF was referenced to the European Qualifications Framework (EQF) in 2014.

Quality assurance and accreditation of institutions and programmes

The Norwegian Agency for Quality Assurance in Education (NOKUT) is an autonomous governmental agency which provides external supervision and control of the quality of Norwegian higher education, as well as of all tertiary vocational education³.

NOKUT accredits new study programmes, controls the existing ones, and provides a cyclic evaluation of the institutions' quality assurance systems for educational provision.

An accredited higher education institution is granted the right to offer educational provision, without having to apply to NOKUT for specific programme accreditation, in accordance with the authority that its institutional category implies:

- a) Universities may, without further external accreditation, establish study programmes at all levels.
- b) Accredited university colleges have to apply for the accreditation of programmes at master and doctoral levels.
- c) In those fields where specialized university institutions and accredited university colleges have the right to award doctorates or corresponding degrees, they may themselves decide which study programmes and disciplines the institution shall provide.

University colleges without institutional accreditation must apply to NOKUT for accreditation of study programmes at all levels.

Lists of all accredited institutions, as well as of all accredited study programmes at the university colleges without institutional accreditation are available on www.nokut.no

Admission requirements

The Higher Education Entrance Qualification is the successful completion of Norwegian upper secondary education in general studies. The Certificate of Upper Secondary Education and Training (*Vitnemål for videregående opplæring*) is based on 13 years of schooling. The certificate states that the student is eligible for further studies (*generell studiekompetanse*).

Admission may also be gained by means of other qualifications recognized as being on a par with the Higher Education Entrance Qualification, such as recognition of prior learning and work experience.

Some fields of study have additional admission requirements.

Academic credit system

All Norwegian higher education institutions use a system of credits (*studiepoeng*) for measuring study activities considered equivalent to the European Credit Transfer and Accumulation System (ECTS). 60 ECTS credits (*studiepoeng*) are allocated to the workload of a full year of academic study, equivalent to 1500-1800 hours of study. 30 ECTS credits are normally allocated to one semester's full-time study. The academic year normally lasts for 10 months and runs from August to June.

¹ National generic learning outcomes descriptions' levels for the bachelor's, master's and doctoral degrees were defined by the Instructions on the Norwegian Qualifications Framework for Higher Education in 2009.

² Learning outcomes for a specific NQF level show the minimum of what each learner should know, understand and be able to do after completing a learning process.

³ Tertiary vocational education (TVE or *høyere yrkesfaglig utdanning*), level 5 in the NQF/EQF, is provided by *fagskoler*, which are considered as tertiary vocational education institutions. TVE is based on upper secondary education and training or equivalent competence. Courses have a duration from six months to two years. All provisions must be accredited by NOKUT.

Degrees and qualifications

NQF/EQF Level 6: Bachelor (1st cycle)

Bachelor's degree is awarded after three years of full-time study (180 ECTS). Some bachelor's degrees, in the field of music and performing arts, consist of four-year bachelor's programmes (240 ECTS).

University college graduate (*høyskolekandidat*) is a two-year degree (120 ECTS), a short cycle degree within the first cycle. Holders of this degree may in some cases continue their studies in a bachelor programme and thus obtain a bachelor's degree.

NQF/EQF Level 7: Master (2nd cycle)

Master's degree is normally obtained after two years of study (120 ECTS), following the completion of a bachelor's degree. A master's degree programme includes independent work (normally a thesis) of at least 30 ECTS.

Experience-based master's degree has a scope of 90 or 120 ECTS, including independent work / thesis of at least 20 ECTS.

Integrated master's degree is a five-year study programme (300 ECTS) which results in a master's degree, with no intermediate bachelor's degree. An exception is the Master of Architecture programme at the Oslo School of Architecture and Design, which has a scope of 330 ECTS.

In the fields of medicine, psychology and theology, **professionally oriented degrees/qualifications** of six years' duration (360 ECTS) are awarded; in the field of veterinary science - after 5 ½-6 years. They have retained the title/degree *candidata/candidatus* from the former degree system.

NQF/EQF Level 8: Doctoral degree (3rd cycle)

Philosophiae doctor (*ph.d.*) is awarded after three years of study, following the completion of a master's degree or a five to six-year professionally oriented degree/qualification.

Philosophiae doctor in artistic research

(*ph.d. i kunstnerisk utviklingsarbeid*) is awarded after three years of study in the field of creative and performing arts. In 2018, the degree replaced the former *Diploma, artistic development programme*, which will be phased out by 2025.

Doctor philosophiae (*dr. philos.*) is conferred on graduates who have qualified for a doctoral degree on their own, without formal research training.

Descriptions of the educational qualifications are given in the Norwegian Qualifications Framework for Lifelong Learning at www.nokut.no/nkr

The Norwegian Education System

